Adam Gatherer

Systems Engineer | Python Automation & AWS Infrastructure

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Profile

Systems and automation engineer with production experience in banking and healthcare. Develops Python tooling, delivers infrastructure changes and investigates dependencies across complex data-processing systems. Combines Linux and Windows Server operations with security remediation and controlled delivery. Builds AWS serverless applications with Python and Terraform through personal projects, including BirdAlert's event-driven ingestion and query platform.

Technical skills

Automation and development: Python, Bash, PowerShell, SQL/SQLite, pandas, Git.

Infrastructure and operations: Linux, Windows Server, VMware vSphere, Nagios, workload orchestration, migrations, incident investigation and change control.

Engineering delivery: Jenkins/CloudBees deployments, GitHub Enterprise, Snyk, code review, ServiceNow.

Cloud & Infrastructure as Code (Projects): AWS serverless, Terraform, event-driven processing, DynamoDB, IAM, CloudWatch, Python unit tests and GitHub Actions.

Employment

Data Engineer - Orchestration

Barclays PLC (formerly Tesco Bank) | Aug 2024 – Present

Support and improve an AutoSys estate of approximately 11,990 jobs and 2,702 workflow boxes across test, pre-production and production.

- Built **Python/SQLite tooling** to parse job definitions, trace dependencies and combine configuration with runtime data, producing repeatable reports for migration planning and decommissioning assessments.
- Implemented orchestration changes for a SAS infrastructure migration covering **1,796 jobs and 535 boxes**, including environment rollouts, shakedown testing, troubleshooting and cutover support.
- Reduced **medium-risk Snyk findings from 62 to 11** in orchestration tooling by addressing unsafe deserialisation, SQL injection and path traversal; improved maintainability through linting and formatting.
- Implemented and supported production data-processing workflows linking ETL, managed file transfer and SAS; investigated failures across scripts, file triggers and scheduling dependencies with partner teams.
- Delivered controlled removal of obsolete workloads across test, pre-production and production, using dependency checks, rollback definitions, approved deployments and post-change validation.

Server Engineer

NHS Lothian | Oct 2022 – Aug 2024

- Delivered the annual server access review with security colleagues and the server team lead; investigated permissions, agreed access requirements with users and automated parts of the review using PowerShell and Python.
- Automated Nagios client installation with PowerShell to reduce manual server-build work; monitored storage and services and responded to capacity and availability issues.
- Deployed and administered VMware vSphere infrastructure and Windows Server systems, including SQL databases for third-party applications.
- Planned and carried out network-directory data migrations, managing permissions, storage requirements and expected growth; arranged controlled server access for software suppliers.

Selected personal projects

BirdAlert | Python, Terraform, AWS

Live beta: birdalert.gatherer.tech | **Source:** GitHub — BirdAlert

AWS serverless application for collecting, processing and querying UK wildlife sightings. Phase 1 implemented and validated in a development environment in October 2026.

- Implemented an ingestion pipeline in Python and Terraform: EventBridge schedules Lambda collection, S3 preserves raw eBird responses and a separate processor validates sightings for DynamoDB.
- Used stable sighting identities and conditional transactional writes to make repeated processing idempotent. Excluded private sightings and retained explicit validation status for provisional records.
- Built an indexed read API with region/species/date filters, cursor pagination and UK daylight-saving support; CloudFront serves the static client.
- Added scoped IAM, Parameter Store, bounded retries, SQS dead-letter queues and CloudWatch alarms. Development checks covered ingestion and processor failure handling; Python tests cover processing and the API.

Stormwatcher | Python, Terraform, AWS

github.com/adam-gatherer/stormwatcher

Built a weather-ingestion and risk-scoring pipeline: scheduled collection lands raw forecast data in S3, a second Lambda validates and transforms it, and DynamoDB stores the results. Implemented configurable SNS status and threshold notifications, with infrastructure defined in Terraform.

JIL Estate Sync | AWS, Terraform

AWS event-driven proof of concept for moving generated scheduler JIL snapshots through S3, EventBridge and CodeBuild into GitHub pull requests, with Terraform-managed IAM. Source

Cloud CV | AWS, GitHub Actions

github.com/adam-gatherer/cloud-cv-aws

Built a serverless CV website following the Cloud Resume Challenge, with S3/CloudFront hosting, a Python Lambda/DynamoDB visitor counter and a GitHub Actions workflow for site synchronisation and cache invalidation.

Terraform Load Balancer | AWS, Terraform

Tutorial-derived IaC project provisioning a VPC, subnets, NAT gateway, application load balancer and Auto Scaling Group. Source

Earlier experience

Systems & Infrastructure Technician

Edinburgh Napier University | Jun 2021 – Oct 2022

- Packaged and troubleshot VMware/Cloudpaging applications, scripted software installation with PowerShell, Bash and Python, and supported lab upgrades with student volunteers.

Customer Service Advisor | Centrica | Jan 2018 – Aug 2019

Customer service and gas/electrical emergency-line operations; automated routine tasks with Windows Task Scheduler.

Education and awards

BEng Computer Systems and Networks

Edinburgh Napier University | Sep 2021 – Jul 2023

Graduated with Distinction · Awarded the Edinburgh Napier University Medal

Study included Python, Linux systems, network architecture, virtualisation and security.

HND Computer Networking — A grade

Edinburgh College | Aug 2019 – Jul 2021

Certifications

Microsoft Certified: Azure Fundamentals; Microsoft Certified: Security, Compliance, and Identity Fundamentals; Microsoft 365 Certified: Fundamentals.